

Mitigating Identity Essentialism in LLM Agents with Longitudinal Life Trajectories

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Abstract

Large language models (LLMs) offer a scalable approach to social simulation, but their credibility depends on how agents are constructed. Existing methods can partially reproduce population-level patterns, yet often fail to capture human-like diversity. Our analysis shows that static-profile agents exhibit stronger demographic separation and within-group compression than humans, a pattern consistent with identity essentialism: *demographic labels can encourage models to treat group-average tendencies as individual traits*, homogenizing responses within groups. We argue that this limitation arises from two related factors: sparse, static agent representations and the limited ability of prompt-only memory to persistently integrate experience. Inspired by complementary memory systems, we propose LifeMem, a longitudinal memory framework that combines structured life-event retrieval with agent-specific parametric memory for experience integration. Experiments on Add Health and Understanding Society with three LLMs show that LifeMem improves alignment with human data in terms of response distributions, overall and within-group diversity, and patterns of within-person response change across life stages. These findings highlight the value of longitudinal life-event memory for constructing more faithful and dynamically evolving social agents.

Code — <https://github.com/halsayxi/LifeMem>

Introduction

Large language models (LLMs) provide a scalable foundation for social simulation, enabling researchers to study human opinions, attitudes, behaviors, and interactions when conventional surveys or experiments are costly or difficult (Aher, Arriaga, and Kalai 2023; Argyle et al. 2023; Park et al. 2023). The credibility of such simulations depends critically on how agents are constructed.

Most approaches instantiate agents from demographic labels, personas, or static profiles (Argyle et al. 2023; Santurkar et al. 2023; Bisbee et al. 2024; Hu and Collier 2024). Although these representations effectively condition models on individual attributes, they often produce responses that are less diverse than human data (Bisbee et al. 2024; Wang, Morgenstern, and Dickerson 2025; Xie et al. 2026; Wang et al. 2026). This diversity collapse may reflect a form of **identity**

essentialism (Wang, Morgenstern, and Dickerson 2025). In social psychology, essentialism treats social categories as reflecting stable underlying essences, making members of the same category appear fundamentally alike (Prentice and Miller 2007; Bastian and Haslam 2006). In social simulation, a corresponding pattern arises when static identity labels become overly predictive of agent responses, compressing within-group variation while amplifying between-group differences. Figure 1 presents a response-space analysis consistent with this pattern. Such distortions may exaggerate demographic differences, reinforce stereotypes, and bias conclusions drawn from simulated populations.

We attribute this limitation to both the content and form of agent memory. First, demographic attributes provide only a sparse and static account of an individual, omitting the richer, evolving experiences associated with between-person differences and within-person change (Elder Jr 1998; Curran and Bauer 2011). Second, prompt-based conditioning may lead to diversity collapse (Wang et al. 2026). As shown in Figure 3, explicit memory alone does not ensure that accumulated experience is persistently integrated into the agent.

Inspired by cognitive science, we propose **LifeMem**, a longitudinal memory framework for constructing dynamically evolving LLM social agents. Human memory is commonly understood as relying on complementary systems: the hippocampal system rapidly preserves individual experiences, whereas the neocortex gradually integrates information across experiences into persistent representations (McClelland, McNaughton, and O’Reilly 1995; Kumaran, Hassabis, and McClelland 2016). LifeMem translates this distinction into two complementary components for representing life experiences: (1) structured life-event memory, which explicitly preserves event content, timing, and supporting evidence for retrieval; and (2) agent-specific LoRA adapters, which encode accumulated experiences as parametric memory persistent across questions and updated over time. By combining richer longitudinal information with persistent parametric memory, LifeMem targets limitations in both the content and representation of agent memory, aiming to construct agents that more faithfully reflect individual differences and temporal change.

We evaluate LifeMem on Add Health (Harris et al. 2019) and Understanding Society (Buck and McFall 2012) us-

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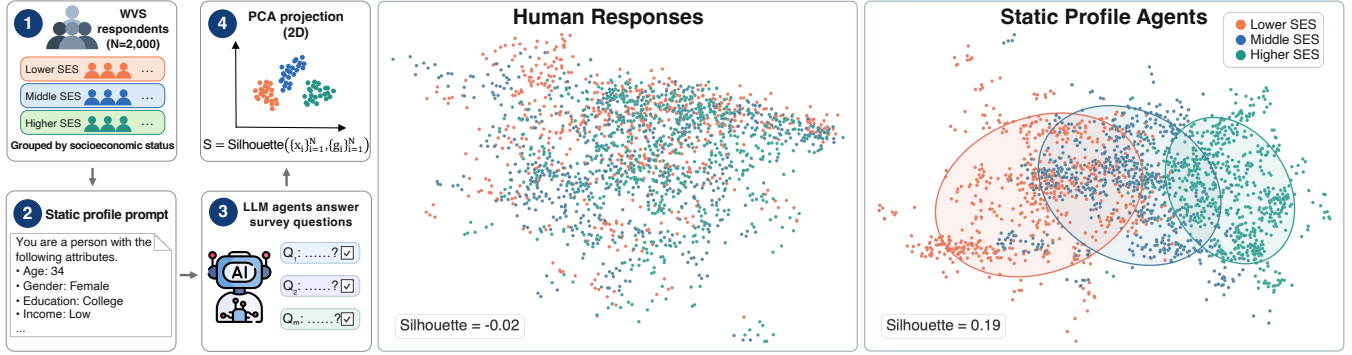


Figure 1: Static demographic conditioning produces a pattern consistent with identity essentialism. We divide 2,000 randomly sampled WVS wave 7 respondents (Haerpfer et al. 2022) into three socioeconomic status groups and compare human responses with those of profile-conditioned Llama-8B agents. Each individual’s responses are concatenated and projected into a two-dimensional PCA space. The separate projections show greater within-group variation and overlap among humans (silhouette -0.02), whereas agents exhibit stronger within-group compression and between-group separation (silhouette 0.19). Silhouette scores are computed in the original response space before PCA projection.

ing three instruction-tuned LLMs from different families: **Llama-3.1-8B-Instruct** (hereafter, **Llama-8B**) (Grattafiori et al. 2024), **Ministral-3-8B-Instruct-2512** (Liu et al. 2026), and **Qwen3.5-9B** (Qwen Team 2026). Compared to static conditioning, diversity-oriented prompting, non-parametric memory, and control baselines, LifeMem improves alignment with human data in terms of response distributions, overall and within-group diversity, and within-person response change across life stages. Our contributions are:

- We identify identity essentialism in static-profile LLM agents, which reduces within-group diversity and exaggerates between-group differences.
- We propose LifeMem, combining structured life-event memory with agent-specific parametric memory to model diverse and evolving individuals.
- Across two longitudinal surveys and three LLMs, LifeMem improves alignment with human distributions, diversity, and within-person changes.

Related Work

Current LLM-based social simulation often relies on persona conditioning and static agent representations. LLMs are increasingly used as experimental participants and survey respondents (Aher, Arriaga, and Kalai 2023; Argyle et al. 2023; Santurkar et al. 2023; Bisbee et al. 2024; Hu and Collier 2024; Lutz et al. 2025). Existing approaches often construct agents from interviews, interaction histories, or social-media records (Park et al. 2024; Du et al. 2025; Li and Conrad 2026; Du et al. 2026; Guo et al. 2026b), supporting partial alignment with populations. However, persona conditioning may reduce response variance or introduce bias (Bisbee et al. 2024; Boelaert et al. 2025; Wang, Morgenstern, and Dickerson 2025). Moreover, static profiles can compress individuals into demographic types (Xie et al. 2026; Hu and Collier 2024). This pattern resembles identity essentialism, which attributes stable properties to social categories (Prentice and Miller 2007; Bastian and Haslam

2006), although demographic conditioning is not inherently essentialist (Wang, Morgenstern, and Dickerson 2025). More broadly, LLM social agents are typically fixed rather than updated from temporally ordered observations. In contrast, longitudinal research distinguishes between-person differences from within-person change and uses detailed life-event histories to better characterize individual trajectories (Curran and Bauer 2011; Elder Jr 1998).

Prior work improves diversity at the output, prompt, memory, or parameter level. Output-level methods modify sampling, decoding (Holtzman et al. 2019; Platt et al. 1999; Li et al. 2016; Vijayakumar et al. 2018; Chung, Kamar, and Amershi 2023; Wong et al. 2026; Zhang et al. 2025), or human involvement (Abels and Lenaerts 2025) to increase response diversity, but the resulting variation is often stochastic rather than grounded in individual experience. Prompt-level methods introduce multilingual, anti-stereotype, biography-based, population-aligned, or ensemble personas (Wang et al. 2025; Sivakumar et al. 2025; Lutz et al. 2025; Ge et al. 2024; Hu et al. 2025; Ashkinaze et al. 2025), but they generally lack an evolving representation of individuals. Memory methods store, retrieve, update, or forget experience (Maharana et al. 2024; Lewis et al. 2020; Park et al. 2023; Zhong et al. 2024; Long et al. 2026; Guo et al. 2026a), but retrieval alone may not support persistent integration. Parameter-level methods encode knowledge or identity into model parameters or activations (Su et al. 2025; Chen et al. 2025; Wang et al. 2026), but are not typically updated from longitudinal observations.

Methodology

This section formulates the longitudinal simulation task and introduces LifeMem’s dual-memory framework. We then describe its structured memory for explicit event retrieval and parametric memory for integrating experiences across waves.

Problem Formulation

Let $\mathcal{I} = \{1, \dots, N\}$ denote a population of individuals and $\mathcal{W} = \{1, \dots, T\}$ the ordered longitudinal waves. Each indi-

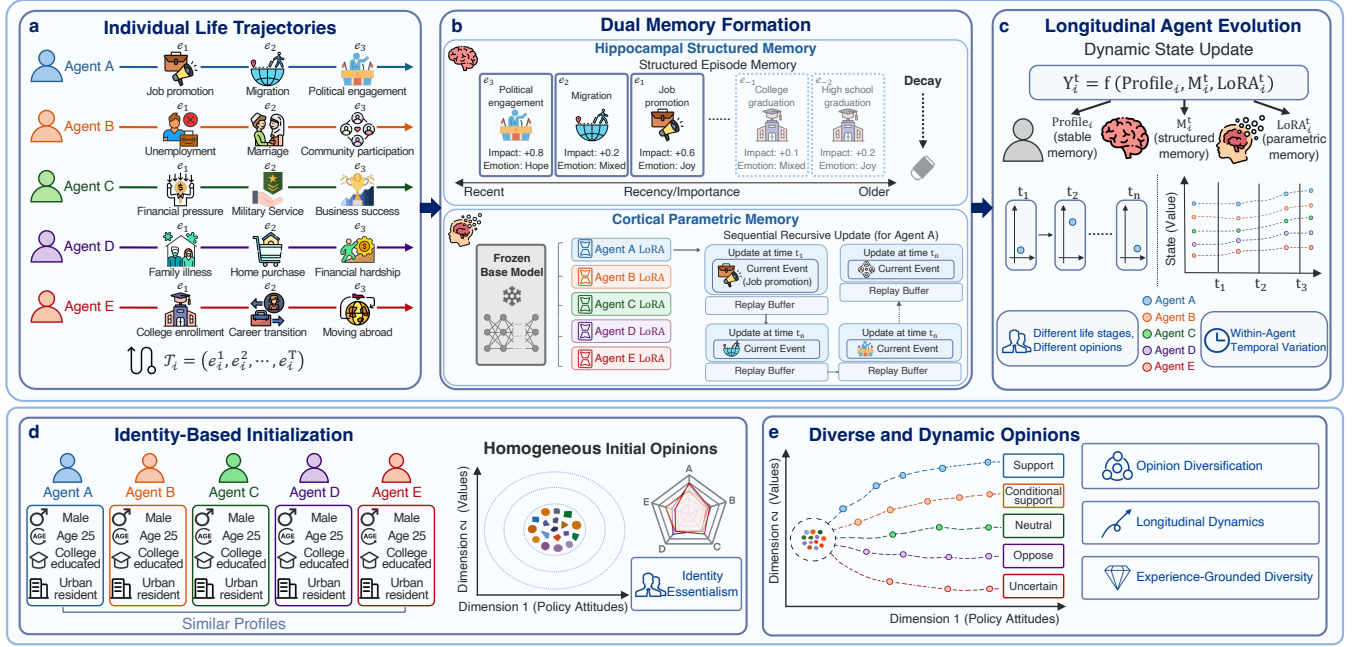


Figure 2: Overview of LifeMem. (a) Individual life trajectories contain diverse longitudinal experiences. (b) Structured memory supports explicit retrieval, while agent-specific LoRA adapters consolidate experiences into parametric memory through recursive updates. (c) The resulting agent state evolves across life stages. (d) Agents are initialized from similar demographic profiles, which may produce homogeneous opinions. (e) LifeMem produces diverse and dynamic opinions.

vidual i has a static profile P_i constructed from background information and a sequence of life events observed at wave t ,

$$\mathcal{E}_{i,t} = \{e_{i,t,k}\}_{k=1}^{K_{i,t}}, \quad (1)$$

where $e_{i,t,k}$ denotes the k -th observed life event and $K_{i,t} = |\mathcal{E}_{i,t}|$. The accumulated trajectory through wave t is

$$\mathcal{T}_{i,\leq t} = \bigcup_{\tau=1}^t \mathcal{E}_{i,\tau}. \quad (2)$$

For an evaluation question $q_{t,j}$, LifeMem predicts

$$\hat{y}_{i,t,j} = F_{\theta, A_{i,t}}(P_i, R_{i,t}(q_{t,j}), q_{t,j}), \quad (3)$$

where θ is the frozen base LLM, $A_{i,t}$ is an agent-specific LoRA adapter, and $R_{i,t}(q_{t,j})$ contains events retrieved from structured memory. LifeMem combines question-dependent evidence with a persistent individual state to model between-person differences and within-person change.

LifeMem Overview

LifeMem is a dual-memory framework inspired by complementary learning systems theory, which distinguishes rapid encoding of specific experiences in the hippocampal system from gradual integration in the neocortex (McClelland, McNaughton, and O'Reilly 1995; O'Reilly and Norman 2002; O'Reilly et al. 2014). In LifeMem, structured memory stores explicit and traceable life events, while agent-specific LoRA adapters integrate accumulated experiences into persistent parametric memory, as illustrated in Figure 2.

Both memories receive the same longitudinal event stream but serve complementary roles. Structured memory provides relevant evidence for the question, whereas parametric memory integrates accumulated experience across questions and waves without repeatedly adding the full trajectory into the prompt. Temporal decay prioritizes retrieval, and replay preserves earlier information during sequential adapter updates.

Hippocampal Structured Memory

Each life event $e_{i,t,k}$ is derived from a survey question-answer pair and converted into a second-person statement $x_{i,t,k}$ while retaining its original question and response. A frozen encoder g_ϕ maps the statement to an embedding

$$\mathbf{h}_{i,t,k} = g_\phi(x_{i,t,k}). \quad (4)$$

The structured memory stores each event together with its embedding and wave index:

$$\mathcal{M}_{i,t}^S = \{(e_{i,\tau,k}, \mathbf{h}_{i,\tau,k}, \tau) \mid 1 \leq \tau \leq t\}. \quad (5)$$

New events are appended without removing earlier records, preserving a traceable account of the observed trajectory.

For a question q , retrieval combines semantic relevance with temporal decay:

$$\text{Score}_t(q, e_{i,\tau,k}) = \text{sim}(g_\phi(q), \mathbf{h}_{i,\tau,k}) \exp[-\lambda(t-\tau)], \quad (6)$$

where $\lambda \geq 0$ controls the preference for recent events. This decay modifies retrieval priority rather than deleting older experiences, which remain available when sufficiently relevant. The top- K events form the retrieved evidence:

$$R_{i,t}(q) = \text{Top-}K_{e \in \mathcal{M}_{i,t}^S} \text{Score}_t(q, e). \quad (7)$$

The selected statements are inserted into the inference prompt, providing question-dependent evidence without exposing the complete history.

Cortical Parametric Memory

LifeMem maintains an independently updated LoRA adapter for each agent while keeping the base LLM frozen. For a target layer l , the agent-specific weight is

$$\mathbf{W}_{i,t}^{(l)} = \mathbf{W}_0^{(l)} + \frac{\alpha_{\text{LoRA}}}{r} \mathbf{B}_{i,t}^{(l)} \mathbf{A}_{i,t}^{(l)}, \quad (8)$$

where r and α_{LoRA} denote the LoRA rank and scaling coefficient. We use $\Theta_{i,t}^{\text{LoRA}}$ to denote all agent-specific LoRA parameters, including $\{\mathbf{A}_{i,t}^{(l)}, \mathbf{B}_{i,t}^{(l)}\}_l$. Separate adapters allow individuals with similar demographic profiles to develop different parametric states as their trajectories diverge.

Each life event yields two training instances: a survey question-answer pair preserving the observed response and an event-reconstruction instance linking the event to the agent’s first-person representation. Together, they encode the survey signal and individualized semantics.

Adapters are updated sequentially across waves:

$$\Theta_{i,t}^{\text{LoRA},(0)} = \Theta_{i,t-1}^{\text{LoRA}}, \quad t > 1, \quad (9)$$

with the first wave initialized from a shared adapter state. At each wave, training combines instances from current events with a small sample of historical instances sampled uniformly at random for replay:

$$\mathcal{L}_{i,t} = \sum_{z \in \mathcal{B}_{i,t}^{\text{cur}}} \ell(z; \theta, \Theta_{i,t}^{\text{LoRA}}) + \eta \sum_{z \in \mathcal{B}_{i,t}^{\text{rep}}} \ell(z; \theta, \Theta_{i,t}^{\text{LoRA}}), \quad (10)$$

where η controls the contribution of replayed experiences. The base model remains frozen, and only the agent-specific low-rank parameters $\Theta_{i,t}^{\text{LoRA}}$ are optimized. Replay serves as a lightweight mechanism for mitigating catastrophic forgetting rather than a separate memory component.

Experimental Setup

Datasets

We evaluate LifeMem on the National Longitudinal Study of Adolescent to Adult Health (Add Health) and the UK Household Longitudinal Study (Understanding Society). We use six waves from Add Health and fifteen waves from Understanding Society. To ensure consistent longitudinal tracking, we retain only respondents observed in all selected waves.

Survey variables are semantically categorized into three types: demographic attributes for initial profiles, life events describing evolving experiences, and evaluation targets. To prevent leakage, evaluation targets are excluded from both demographic and life-event variables. To evaluate within-person opinion changes, we further identify a shared set of evaluation questions available in every wave of Understanding Society and use them as the longitudinal evaluation set.

Baselines

Static conditioning includes **DIRECT** and **PROFILE** (Argyle et al. 2023; Santurkar et al. 2023; Bisbee et al. 2024; Hu and Collier 2024); diversity-oriented prompting includes **MULTILINGUAL** (Wang et al. 2025) and **ANTI-STEREOTYPE** (Sivakumar et al. 2025); and non-parametric memory includes **SIMVBG** (Du et al. 2025), **FULL HISTORY** (Maharana et al. 2024), and **EVENT RAG** (Lewis et al. 2020; Maharana et al. 2024). **RANDOM EVENT** uses random events to test the importance of respondent-specific trajectories.

Evaluation Metrics

We use four lower-is-better metrics. *KL divergence* measures alignment between human and agent response distributions. The *within-group pairwise distance gap* is the absolute difference between human and agent within-group pairwise response distances. A smaller gap indicates that simulated within-group diversity is closer to that of humans, suggesting less identity-essentialist behavior. The *normalized entropy gap* is the absolute difference between the normalized response entropies of humans and agents. The *transition-distribution JS divergence* measures how closely the model reproduces the human population-level distribution of response transitions between adjacent survey waves.

Implementation Details

We use three similarly sized LLMs: Llama-3.1-8B-Instruct, Ministral-3-8B-Instruct-2512, and Qwen3.5-9B. Using a random seed of 42, we sample 100 respondents per dataset. All methods share the same samples, questions, backbones, and deterministic decoding settings, with temperature 0 and sampling disabled. **EVENT RAG** uses the stronger bge-m3 retriever, whereas LifeMem uses the lightweight all-MiniLM-L6-v2 encoder. Both methods retrieve the top- $K = 5$ events. LifeMem uses a temporal decay factor of 0.105, rank-8 LoRA adapters, replay size 4, and replay weight 0.5. Experiments run on one NVIDIA A800-SXM4 80 GB GPU.

Experiments and Results

This section evaluates LifeMem’s effectiveness, mechanisms, scaling, and efficiency. We examine explicit-memory limitations, overall performance, parameter differentiation, component contributions, and computational costs.

Limits of Explicit Prompt Memory

We test whether retrieving more explicit life-history information is sufficient for longitudinal simulation by varying **EVENT RAG** retrieval depth from $K = 1$ to $K = 180$ (i.e., **FULL HISTORY**) on Add Health with Llama-8B. We evaluate distributional and diversity alignment together with runtime and average input-token use to characterize both effectiveness and computational cost.

As shown in Figure 3, the three simulation metrics initially improve but largely saturate around $K = 40$ and sometimes deteriorate, while runtime and input-token use continue to increase. This trade-off illustrates a practical limitation of prompt-only memory and motivates combining selective retrieval with persistent parametric memory.

Category	Method	Add Health			Understanding Society			
		KL Div. ↓	WG Gap ↓	Ent. Gap ↓	KL Div. ↓	WG Gap ↓	Ent. Gap ↓	Trans. JS ↓
LLAMA-3.1-8B-INSTRUCT								
Static	Direct	14.7551*	0.5596*	0.7053*	15.3734*	0.5561*	0.7076*	0.4580*
Conditioning	Profile	8.6719*	0.3951*	0.5112*	7.1522*	0.3835*	0.4972*	0.3886*
Diversity-Oriented	Multilingual	10.4654*	0.3733*	0.5035*	11.7367*	0.3446*	0.4763*	0.4570*
Prompting	Anti-Stereotype	9.8270*	0.4163*	0.5452*	8.1485*	0.4066*	0.5120*	0.3592*
Non-Parametric Memory	SimVBG	6.8058*	0.3380*	0.4345*	6.2912*	0.3544*	0.4730*	0.3709*
	Full History	6.1176*	0.3079*	0.3974*	4.9344*	0.3198*	0.4004*	0.3633*
	Event RAG	5.8264*	0.2964*	0.3951*	5.2935*	0.3163*	0.3991*	0.3648*
Control Baseline	Random Event	6.8821*	0.3524*	0.4538*	5.5673*	0.3429*	0.4539*	0.3803*
Proposed Method	LifeMem	4.0635	0.2309	0.3207	3.4529	0.2886	0.3742	0.3331
MINISTRAL-3-8B-INSTRUCT-2512								
Static	Direct	15.5952*	0.5679*	0.7161*	15.4925*	0.5648*	0.7182*	0.5076*
Conditioning	Profile	8.8060*	0.4120*	0.5263*	6.8298*	0.3622*	0.4622*	0.3609
Diversity-Oriented	Multilingual	12.7028*	0.3597*	0.5129*	13.1824*	0.3942*	0.5350*	0.4889*
Prompting	Anti-Stereotype	10.0458*	0.4339*	0.5537*	8.7660*	0.3946*	0.5066*	0.3757
Non-Parametric Memory	SimVBG	6.9946*	0.3427*	0.4566*	4.8323*	0.3272*	0.4156*	0.3610
	Full History	4.7175*	0.2893*	0.3817*	3.8053*	0.2927*	0.3413*	0.3528
	Event RAG	5.4368*	0.2849*	0.3895*	4.3342*	0.2887*	0.3413*	0.4693*
Control Baseline	Random Event	6.1787*	0.3578*	0.4647*	4.7275*	0.3156*	0.4048*	0.3924
Proposed Method	LifeMem	2.2959	0.1928	0.2783	1.9659	0.2277	0.2659	0.3399
QWEN3.5-9B								
Static	Direct	13.9957*	0.5482*	0.6923*	15.9998*	0.5559*	0.7080*	0.4452*
Conditioning	Profile	8.0174*	0.4004*	0.5093*	5.3214*	0.3463*	0.4325*	0.3471*
Diversity-Oriented	Multilingual	7.7194*	0.2211	0.3395	10.9903*	0.3438*	0.4728*	0.4405*
Prompting	Anti-Stereotype	7.7407*	0.3743*	0.4770*	6.0185*	0.3422*	0.4254*	0.3356*
Non-Parametric Memory	SimVBG	5.7090*	0.3307*	0.4311*	4.4864*	0.3182*	0.3889*	0.3298*
	Full History	4.5266	0.2599	0.3432	3.7659*	0.2979*	0.3441*	0.3203*
	Event RAG	4.3701	0.2676	0.3574	3.5507*	0.2673*	0.3070	0.3394*
Control Baseline	Random Event	6.0741*	0.3625*	0.4584*	3.7848*	0.3031*	0.3707*	0.3466*
Proposed Method	LifeMem	4.1177	0.2723	0.3601	2.6879	0.2519	0.2932	0.3060

Table 1: Overall results on Add Health and Understanding Society across three models. KL Div., WG Gap, Ent. Gap, and Trans. JS denote KL divergence, within-group pairwise distance gap, normalized entropy gap, and transition-distribution JS divergence. Lower values are better for all metrics. Bold marks the best result for each model and metric; LifeMem is shaded light gray. * indicates results significantly worse than LifeMem in a two-sided paired t -test conducted over shared questions ($p < 0.05$).

Overall Performance and Diversity

Table 1 shows that LifeMem performs consistently well across four dimensions, with largely significant gains across three backbones and two datasets. It reduces distributional and diversity gaps relative to human responses across settings. DIRECT performs similarly poorly across models, while Ministral shows the largest improvement with LifeMem.

PROFILE improves over DIRECT but remains behind LifeMem, suggesting that static demographic profiles cannot fully capture human heterogeneity and temporal variation in these settings. MULTILINGUAL and ANTI-STEREOTYPE prompting yield inconsistent gains, suggesting that these prompting strategies do not reliably reproduce human-like diversity across settings. SIMVBG, FULL HISTORY, and EVENT RAG perform better than static baselines but still generally trail LifeMem, suggesting that non-parametric memory alone may be insufficient to achieve the same level of alignment. LifeMem also outperforms RANDOM EVENT, highlighting the importance of respondent-specific trajectories.

One notable exception is Qwen3.5-9B on Add Health, where MULTILINGUAL achieves slightly lower within-group pairwise distance gap and normalized entropy gap than LifeMem. However, its substantially higher KL divergence indicates that these gains in diversity alignment are accompanied by poorer agreement with the overall human response distribution, suggesting a less balanced improvement across evaluation dimensions.

Overall, LifeMem improves alignment with human data across response distributions, overall and within-group diversity, and longitudinal response dynamics. These results suggest that LifeMem mitigates patterns consistent with identity essentialism by reducing within-group homogenization while maintaining closer alignment with human responses.

Ablation Study

Table 2 evaluates the contributions of structured and parametric memory. Removing either component significantly degrades performance on most metrics, supporting their

Method	Add Health			Understanding Society			
	KL Div. ↓	WG Gap ↓	Ent. Gap ↓	KL Div. ↓	WG Gap ↓	Ent. Gap ↓	Trans. JS ↓
LLAMA-3.1-8B-INSTRUCT							
LifeMem (Full)	4.0635	0.2309	0.3207	3.4529	0.2886	0.3742	0.3331
LifeMem w/o Param. Mem.	5.3517*	0.2554*	0.3436	5.0874*	0.3259*	0.4150*	0.3594*
LifeMem w/o Struct. Mem.	6.2815*	0.3132*	0.4164*	4.7555*	0.3379*	0.4318*	0.3614*
MINISTRAL-3-8B-INSTRUCT-2512							
LifeMem (Full)	2.2959	0.1928	0.2783	1.9659	0.2277	0.2659	0.3399
LifeMem w/o Param. Mem.	4.6504*	0.2516*	0.3456*	3.8454*	0.2810*	0.3389*	0.3838
LifeMem w/o Struct. Mem.	3.3799*	0.2627*	0.3597*	2.4410*	0.2483*	0.3093*	0.3443
QWEN3.5-9B							
LifeMem (Full)	4.1177	0.2723	0.3601	2.6879	0.2519	0.2932	0.3060
LifeMem w/o Param. Mem.	4.4295	0.2733	0.3604	3.4224*	0.2738*	0.3193*	0.3411*
LifeMem w/o Struct. Mem.	6.0934*	0.3651*	0.4760*	3.2797*	0.2842*	0.3430*	0.3235*

Table 2: Ablation study of LifeMem on Add Health and Understanding Society. KL Div., WG Gap, Ent. Gap, and Trans. JS denote KL divergence, within-group pairwise distance gap, normalized entropy gap, and transition-distribution JS divergence. All metrics are lower-is-better. Bold marks the best result for each model and metric; LifeMem (Full) is shaded light gray. * indicates results significantly worse than LifeMem (Full) in a two-sided paired t -test conducted over shared questions ($p < 0.05$).

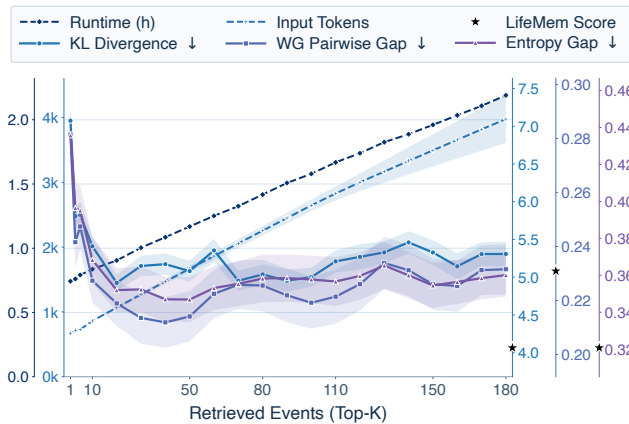


Figure 3: Effect of EVENT RAG retrieval depth on Add Health with Llama-8B across 100 agents. We report alignment metrics, runtime, and input tokens for top- $K \in [1, 180]$. Stars mark LifeMem at $K = 5$. The shaded bands indicate standard deviation across waves.

complementarity. Parametric memory maintains a persistent agent state that integrates experiences across waves beyond the events retrieved for the current question, whereas structured memory supplies explicit, traceable, and temporally grounded evidence for generation and tends to contribute more strongly to diversity alignment. Their relative effects vary across different models and datasets.

Evolution of Agent-Specific LoRA States

Figure 4 uses PCA to project the high-dimensional agent-specific LoRA states into a shared two-dimensional space, illustrating how the states of 100 Llama-8B agents evolve

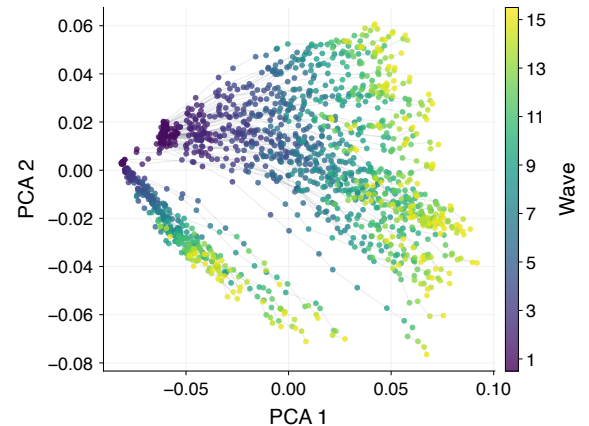


Figure 4: PCA visualization of LifeMem’s agent-specific LoRA states for 100 Understanding Society agents using Llama-8B. Points represent states after each wave, with trajectories connecting the same agent over time. Agent states increasingly diverge as life histories accumulate.

over 15 waves on Understanding Society. The initially concentrated adapters become progressively more dispersed and follow distinct trajectories as individual histories accumulate over time. This pattern suggests that sequential updates gradually encode heterogeneous life experiences into differentiated parametric states, reflecting increasing personalization across agents with different longitudinal trajectories.

Scaling with Accumulated Life Experiences

Figure 5 examines the effect of longitudinal life-event coverage in LifeMem on the Add Health dataset with Llama-8B. We vary the maximum number of available events per wave

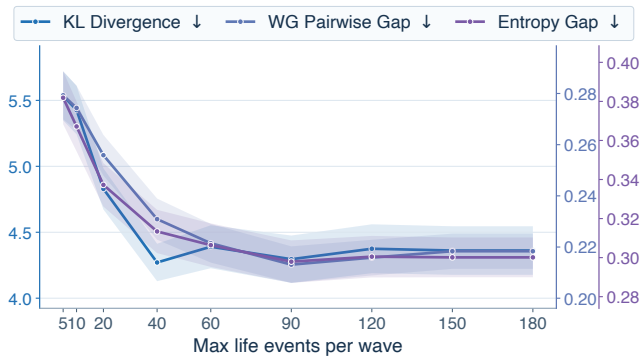


Figure 5: Effect of life-event coverage in LifeMem on Add Health with Llama-8B. We simulate 100 agents while varying the maximum number of life events available per wave from 5 to 180. Curves report KL divergence, within-group pairwise distance gap, and normalized entropy gap. The shaded bands indicate standard deviation across waves. Increasing life-event coverage generally improves alignment with human response distributions and diversity.

from 5 to 180 for 100 agents. Increasing coverage generally reduces KL divergence, normalized entropy gap, and within-group pairwise distance gap, with the improvements gradually plateauing after roughly 90 events per wave. This pattern indicates that richer longitudinal histories provide more informative and individualized evidence for distinguishing individuals and recovering more human-like response distributions and levels of diversity over time. Overall, these results further support our motivation to model agents through accumulated life experiences rather than relying on static demographic profiles alone.

Efficiency Analysis

All efficiency measurements are obtained under the same hardware configuration. We focus primarily on relative trends and comparisons across methods, as absolute runtime may vary with different hardware characteristics, including CPU, GPU, memory, and storage configurations.

Online inference. Table 3 compares the average per-question inference latency across different methods on the Add Health and Understanding Society datasets using Llama-8B. Retrieval latency is included for both the EVENT RAG and LifeMem methods to ensure a consistent end-to-end comparison. LifeMem is slower than lightweight prompt-only baselines but remains faster than FULL HISTORY, EVENT RAG, and SIMVBG on both datasets. Adapter loading introduces little additional latency and can be amortized across questions once an agent’s adapter is loaded.

Storage. Figure 6 compares per-agent adapter and structured life-event storage for Llama-8B as longitudinal histories accumulate. Structured event storage grows with trajectory length because newly observed experiences must be retained explicitly, whereas each agent maintains a single fixed-size LoRA adapter that is updated after each wave. The updated adapter is then used for subsequent inference, so parametric-memory storage remains bounded rather than growing with

Method	AH		USoc	
	Time	Rel.	Time	Rel.
Direct	13.8	1.0	14.4	1.0
Profile	65.2	4.7	70.8	4.9
Multilingual	24.5	1.8	26.5	1.8
Anti-Stereotype	66.8	4.9	72.9	5.1
SimVBG	196.4	14.3	196.4	13.7
Full History	334.5	24.3	318.1	22.1
Event RAG	1094.2	79.5	956.2	66.5
Random Event	73.0	5.3	79.8	5.6
LifeMem	161.7 _{+0.9}	11.7	154.5 _{+1.4}	10.7

Table 3: Average per-question inference latency with Llama-8B. AH and USoc denote Add Health and Understanding Society. Time is reported in milliseconds, and Rel. denotes latency relative to DIRECT. Retrieval latency is included for EVENT RAG and LifeMem. For LifeMem, the subscripted $+x$ indicates the LoRA adapter-loading latency. Rel. excludes the one-time LoRA adapter-loading latency.

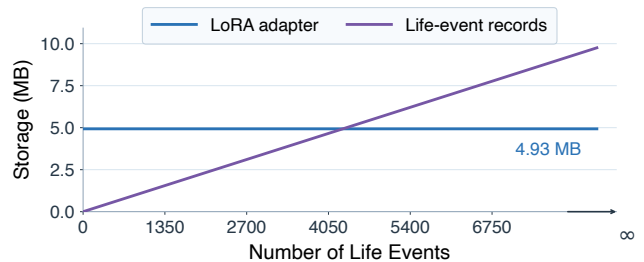


Figure 6: Per-agent storage with Llama-8B as life events accumulate. The fixed-size LoRA adapter remains constant, while structured life-event storage grows with trajectory length.

the number of accumulated events.

Offline training. With Llama-8B, the average offline adapter update time per agent and wave is 7.21 seconds on Add Health and 2.63 seconds on Understanding Society. This one-time cost is incurred during wave-level updates and is not repeated for each subsequent online query.

Conclusion

This work aims to mitigate patterns consistent with identity essentialism in LLM social simulation, where static profiles can suppress individual diversity and temporal change. LifeMem combines structured life-event retrieval with agent-specific LoRA parametric memory. Across two longitudinal datasets and three model backbones, it consistently improves alignment with human response distributions, diversity, and changes across life stages. Overall, the results suggest that, in these settings, incorporating longitudinal life experiences together with persistent parametric memory provides a more faithful representation of individuals.

Ethical Statement

We use only public-use, de-identified survey data, comply with applicable data-use agreements, avoid re-identification, and report aggregate results. Because agent-specific adapters may memorize respondent information, we do not release respondent-level adapters. These agents are not digital replicas and should not be used for consequential person-level decisions; they are intended for methodological research on longitudinal LLM social simulation.

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Limitations

LifeMem is limited by the coverage and granularity of longitudinal surveys, which cannot fully capture an individual’s life experiences. Surveys may omit events outside the questionnaire, record them only at coarse intervals, and fail to capture event duration or respondents’ subjective interpretations. Thus, reconstructed trajectories remain partial representations of personal development rather than complete life histories. Moreover, self-reported survey responses may be affected by recall bias and question framing, and do not necessarily correspond to real-world behavior.

Details of the Motivating Analyses

WVS Static-Profile Experiment

Experimental setup. We use Wave 7 of the World Values Survey (WVS) (Haerpfer et al. 2022) and randomly sample 2,000 respondents with seed 42. For each respondent, we construct a static profile from answers in the WVS demographic section, corresponding to Questions Q260–Q290. The profile is provided to Llama-3.1-8B-Instruct, which is instructed to simulate the corresponding respondent and answer the survey questions. This setup represents a typical profile-conditioned social simulation in which each agent is reconstructed from demographic attributes.

Prompt template. The following prompt template is instantiated with the recoded demographic attributes of each respondent:

Prompt Template

Forget you are an AI model. Simulate a human being. Please answer the following question truthfully.

Please answer based on the following personal profile:

You are <gender>. You are aged <age>. You live in <country>. You are <citizenship status>. You were born <birthplace status>. Your mother was born <mother’s birthplace status>. Your father was born <father’s birthplace status>. You live in <urbanicity>, <settlement type>, with a population of <population size>. Your household consists of <household size> members. You <parent cohabitation status>. You speak <home language> at home. You are <marital status> and <children description>. Your education level is <education level>. Your spouse’s education level is <spouse education level>. You are <employment status>. You work as <occupation>. You work in <employment sector>. Your spouse is <spouse employment status>. Your spouse works as <spouse occupation>. You are <chief wage earner status>. During the past year, your family <household financial change>. You belong to <subjective social class>. Your income is in income group <income group> (1 = lowest, 10 = highest). You identify as <religion>. Your ethnic group is <ethnic group>.

Question: <question>

Options: <options>

Please ONLY output a number. Do NOT write any words, symbols, punctuation, or explanations.

Response processing. For both human respondents and their corresponding agents, the survey answers are converted into response vectors. Each question dimension is standardized before dimensionality reduction, after which PCA is applied to obtain two-dimensional representations. Human and agent responses follow the same response labeling, processing, standardization, PCA, and visualization procedure.

Socioeconomic status construction. We construct a composite socioeconomic status (SES) measure from four WVS variables: subjective social class (Q287), household economic situation (Q286), educational attainment (Q275), and income group (Q288).

Each variable is converted to a directionally consistent numerical scale such that larger values indicate higher socioeconomic status. Let $\tilde{x}_{im}$ denote the recoded value of variable m for respondent i , and let $\mathcal{M}_i$ denote the set of available SES variables for that respondent. The composite score is

$$SES_i = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \tilde{x}_{im}. \quad (11)$$

Respondents are divided into low-, middle-, and high-SES groups according to the tertiles of the composite score.

Silhouette score. We use the silhouette score to quantify how strongly the response representations separate according to the predefined SES groups. Let C_i denote the SES group containing individual i . The average distance between i and other members of the same group is

$$a(i) = \frac{1}{|C_i| - 1} \sum_{\substack{j \in C_i \\ j \neq i}} d(i, j). \quad (12)$$

The minimum average distance between i and any other SES group is

$$b(i) = \min_{C \neq C_i} \frac{1}{|C|} \sum_{j \in C} d(i, j). \quad (13)$$

The silhouette value for individual i is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad (14)$$

and the overall silhouette score is

$$S = \frac{1}{N} \sum_{i=1}^N s(i). \quad (15)$$

Silhouette scores are computed in the standardized response space before PCA projection; PCA is used only for visualization.

A score close to 1 indicates compact within-group representations and clear between-group separation, a score close to 0 indicates substantial overlap, and a negative score indicates that some individuals are closer to another group than to their assigned group.

In this work, the silhouette score is used only to quantify the extent to which response representations align with

predefined demographic groups. It is not interpreted as a standalone measure of simulation quality.

Result summary. Human responses are broadly dispersed and substantially overlap across SES groups, resulting in a silhouette score of $S = -0.02$. In contrast, agent responses form more compact within-group clusters and show clearer separation between SES groups, resulting in $S = 0.19$. This pattern suggests that static demographic prompting may compress complex individuals into group-typical representatives, thereby reducing within-group heterogeneity and amplifying between-group differences. We treat this experiment as a motivating diagnostic that reveals a pattern consistent with identity essentialism in LLM agents.

Event RAG Retrieval-Depth Analysis

We analyze how retrieval depth affects the effectiveness and computational cost of the EVENT RAG baseline. Experiments are conducted on Add Health using Llama-3.1-8B-Instruct and 100 respondents sampled with random seed 42. Event retrieval uses all-MiniLM-L6-v2, consistent with the retrieval component used by LifeMem in this comparison. Its lightweight architecture keeps retrieval overhead low while maintaining competitive retrieval quality.

We vary the retrieval depth (Top- K) from 1 to 180. For each value of K , EVENT RAG retrieves the K highest-ranked life events relevant to each evaluation question and inserts them into the model prompt. We report KL divergence, within-group pairwise distance gap, and normalized entropy gap, where lower values indicate closer agreement with the corresponding human distributions.

Runtime. Runtime is measured as the end-to-end wall-clock duration of one complete EVENT RAG experiment over all 100 respondents and all six survey waves. It therefore includes model loading, population initialization, life-event loading, event retrieval, prompt construction, model inference, output parsing, and metric computation.

Input tokens. For each retrieval depth K , we average the estimated input tokens over all respondent-question prompts across all six survey waves. The reported value therefore represents the mean input length per evaluation prompt.

Comparison with LifeMem. Figure 3 in the main paper additionally marks LifeMem at $K = 5$ as a reference. While increasing EVENT RAG retrieval depth initially improves distributional and diversity alignment, the gains largely plateau around $K = 40$ and sometimes deteriorate, whereas runtime and input-token use continue to increase. Notably, LifeMem already achieves substantially better performance with only $K = 5$, suggesting that its gains are unlikely to be explained solely by retrieving more life events. Instead, these results motivate combining selective retrieval with persistent parametric memory.

Dataset Construction and Preprocessing

Dataset Overviews

Add Health. The National Longitudinal Study of Adolescent to Adult Health (Add Health) follows a nationally

Wave	Respondents	Wave	Respondents
I	6,504	IV	5,114
II	4,834	V	4,196
III	4,882	VI	3,937

Table 4: Number of respondents in the public-use Add Health files across Waves I–VI.

Wave	Respondents	Wave	Respondents
1	77,308	9	52,694
2	77,495	10	50,113
3	70,671	11	46,965
4	65,626	12	43,577
5	61,512	13	41,601
6	64,712	14	55,400
7	59,941	15	48,896
8	56,608		

Table 5: Number of respondents in the public-use Understanding Society files across Waves 1–15.

representative U.S. cohort initially recruited from students in Grades 7–12 during the 1994–1995 school year (Harris et al. 2019). We use all six available survey waves: Wave I (1994–1995), Wave II (1996), Wave III (2001–2002), Wave IV (2008–2009), Wave V (2016–2018), and Wave VI (2022–2025). These waves trace respondents from adolescence through young adulthood and into early midlife.

The survey content evolves with the cohort’s life stage. Waves I–II focus on adolescent health, education, family and peer relationships, and risk behavior. Wave III expands to relationship, fertility, education, and employment histories during the transition to adulthood. Waves IV–VI further examine adult health, socioeconomic circumstances, cognition, caregiving, and family life. Together, these waves provide rich longitudinal records spanning education, employment, health, family, and residential transitions.

The public-use files contain between 3,937 and 6,504 respondents per wave across Waves I–VI. Detailed wave-level sample sizes are reported in Table 4. Respondents are linked across waves using the stable person identifier AID.

Understanding Society. Understanding Society, also known as the UK Household Longitudinal Study (UKHLS), is an annual household panel survey that began in 2009 (Buck and McFall 2012). We use UKHLS Waves 1–15, covering interviews conducted from 2009 to 2024. Each wave is fielded over an overlapping period, while individual sample members are generally interviewed approximately one year apart.

Understanding Society follows members of sampled UK households and covers household composition, family relationships, education, employment, income, housing, health and well-being, finances, political attitudes, and social participation. Its annual panel structure and broad question coverage across fifteen waves support the evaluation of changes in respondents’ answers between adjacent waves.

The public-use files contain between 41,601 and 77,495

respondents per wave across Waves 1–15. Detailed wave-level sample sizes are reported in Table 5. Respondents are linked across waves using the stable person identifier `pidp`.

Respondent Alignment and Sampling

To construct complete longitudinal trajectories, we align respondents using stable cross-wave identifiers and retain only individuals observed in every wave of the corresponding dataset. After alignment, 2,048 Add Health respondents are observed with the same `AID` across all six waves, while 14,104 Understanding Society respondents are observed with the same `pidp` across all fifteen waves.

For the main experiments, we randomly sample 100 respondents from each aligned pool using seed 42. The sampled respondent identifiers are fixed and shared across all methods and backbone models within each dataset. Consequently, all comparisons use the same individuals, survey questions, and observed life histories, thereby controlling for differences in respondent composition. Additional experiments using seeds 43 and 44 evaluate robustness to respondent sampling, as reported in Appendix .

Variable Classification

Structured Question Records. Each selected variable is represented as a structured question record with the following fields:

- **Variable:** the survey variable name.
- **Section:** the questionnaire section.
- **Question:** the natural-language survey question.
- **Options:** the response options for closed-ended questions.
- **Note:** additional constraints for open-ended responses.

Functional Categories. We organize survey variables into three functional categories: *demographic*, *life event*, and *evaluate*. The *demographic* variables are used to construct the initial profile of an agent, including background and profile attributes such as age, gender, education, employment, household composition, region, religion, and family background. The *life event* variables describe longitudinal experiences and state changes, such as schooling, work transitions, partnership changes, fertility, housing changes, health events, caregiving, economic hardship, and other major events. These variables are used to construct structured memory and parametric memory updates across waves. The *evaluate* variables are reserved exclusively for model evaluation and are excluded from profile construction, structured memory, and parametric-memory updates. Examples include questions about residential preferences, internet use, social participation, political attitudes, and subjective well-being.

Classification and Filtering. Variables are classified using a rule-based semantic matching procedure followed by response-quality filtering. The semantic classifier considers the section, question, options, note, and variable fields. We maintain category-specific keyword dictionaries for the three categories. After semantic matching, we filter variables using response-quality statistics computed from the human answer files. For each variable, we compute the total number of responses, the number of valid responses, and the number and

Dataset	Wave	Demo.	Event	Eval.
Add Health	1	73	172	75
	2	65	171	75
	3	76	179	56
	4	65	179	72
	5	70	171	59
	6	75	179	53
UKHLS	1	68	72	11
	2	62	37	11
	3	67	55	38
	4	69	55	37
	5	61	69	49
	6	66	85	41
	7	62	54	26
	8	69	47	16
	9	68	68	34
	10	68	44	28
	11	71	56	50
	12	65	57	64
	13	70	65	25
	14	70	66	43
	15	65	89	45

Table 6: Numbers of demographic, life-event, and evaluation variables selected for each dataset and survey wave. Demo., Event, and Eval. denote demographic, life-event, and evaluation variables, respectively. UKHLS denotes Understanding Society.

rate of missing responses. Variables with severe missingness or too few valid responses are removed. Table 6 reports the number of selected variables in each category for every dataset and wave.

Examples of Classified Variables. The following examples illustrate the three variable categories using questions from Add Health Wave 1.

Demographic Variables

Variable: `BIO_SEX`

Section: Section A: Setup of CAPI Interview

Question: What is your gender?

Options:

- 1: Male;
- 2: Female

Variable: `H1GI1M`

Section: Section 1: General Introductory

Question: Which month were you born in?

Options:

- 1: January;
- 2: February;
- 3: March;
- 4: April;

5: May;
6: June;
7: July;
8: August;
9: September;
10: October;
11: November;
12: December

Life-Event Variables

Variable: H1GH2

Section: Section 3: General Health

Question: How often have you had a headache?

Options:

0: Never;
1: Just a few times;
2: About once a week;
3: Almost every day;
4: Every day

Variable: H1GH43

Section: Section 3: General Health

Question: During the past 30 days, how often did you drive a car or other vehicle when you had been drinking alcohol?

Options:

0: Never;
1: 1 time;
2: 2 or 3 times;
3: 4 or 5 times;
4: 6 or more times

Evaluation Variables

Variable: H1ED19

Section: Section 5: Academics and Education

Question: You feel close to people at your school.

Options:

1: Strongly agree;
2: Agree;
3: Neither agree nor disagree;
4: Disagree;
5: Strongly disagree

Variable: H1PF4

Section: Section 18: Personality and Family

Question: You are satisfied with the way your mother and you communicate with each other.

Options:

1: Strongly agree;
2: Agree;
3: Neither agree nor disagree;
4: Disagree;
5: Strongly disagree

Common Longitudinal Evaluation Set

In addition to the wave-specific evaluation variables, we construct a common longitudinal evaluation set for Understanding Society. This set provides a fixed target space across waves, enabling us to evaluate whether agents reproduce not only population-level response distributions but also changes in the same survey variables over time.

We first retain variables that appear in the question files for all 15 waves, yielding 198 shared variables. We then exclude variables used as demographic or life-event inputs, leaving 136 candidates. After semantic and response-quality filtering, the final evaluation set contains 50 variables covering housing, employment, income, caregiving, family, education, and related life outcomes.

Transition-distribution JS divergence is computed only on Understanding Society because it provides a fixed set of comparable evaluation variables across all 15 waves. Add Health contains fewer waves and substantially less overlap among comparable evaluation variables, so we do not report Transition JS for Add Health.

Shared Experimental Setup

Inference Protocol

We evaluate all methods using three instruction-tuned backbone models: Llama-3.1-8B-Instruct, Ministral-3-8B-Instruct-2512, and Qwen3.5-9B. Thinking mode is disabled for Qwen3.5-9B, and each model uses its native tokenizer and chat template. All methods share the same deterministic decoding configuration, with temperature 0, sampling disabled, a maximum input length of 4,096 tokens, and a maximum output length of 16 tokens. Inference is conducted with a batch size of four using `bfloat16` precision.

All methods also share the same survey-response prompt structure. Method-specific context, such as a demographic profile or retrieved life events, is inserted before the question when applicable. For closed-ended questions, the model is required to return exactly one option number. For open-ended questions, it is instead instructed to provide a concise response without explanation.

Shared Survey Prompt

You are not an AI assistant; you are role-playing a human survey respondent.

Question: <question>

For closed-ended questions:

Options: <options>

Answer with exactly one option number. Do not provide any explanation.

For open-ended questions:

Notes: <notes>

Answer concisely. Do not provide any explanation.

Survey-to-Statement Conversion

We convert each survey question-answer pair into one second-person statement and one first-person statement. The second-person statements are used to construct demographic profiles and structured life-event memory, while the first-person statements serve as reconstruction targets for parametric-memory training.

Conversion is performed using gpt-3.5-turbo-0125, with temperature 0 and a maximum output length of 2,048 tokens. The conversion prompt is shown below.

Survey-to-Statement Conversion Prompt

You convert survey question-answer pairs into concise personal statements.

Rewrite the provided question and answer as one faithful second-person statement and one first-person statement about the respondent.

Rules:

1. Preserve the exact meaning of the answer.

2. Do not add information, explanations, causes, stereotypes, or assumptions.

3. Preserve negation, quantities, dates, frequencies, and time ranges.

4. For life events, do not convert an event into a permanent trait unless the answer says so.

5. Return valid JSON only with keys `second_person` and `first_person`.

Input: <input>

For example, the survey response

Example Survey Response

Question: How often have you had trouble falling asleep or staying asleep?

Answer: Never

is converted into:

Example Converted Statements

Second_Person: You have never had trouble falling asleep or staying asleep.

First_Person: I have never had trouble falling asleep or staying asleep.

This conversion preserves the observed survey response while producing textual representations suitable for profile construction, event retrieval, and adapter training.

Baseline Implementation Details

Static Conditioning

DIRECT. DIRECT (Argyle et al. 2023; Santurkar et al. 2023; Bisbee et al. 2024; Hu and Collier 2024) receives only the shared survey instruction, the current question, its response options or notes, and the corresponding output constraint.

It does not receive demographic information or longitudinal life events.

PROFILE. PROFILE (Argyle et al. 2023; Santurkar et al. 2023; Bisbee et al. 2024; Hu and Collier 2024) additionally conditions the model on each respondent’s fixed initial demographic profile.

Profile Conditioning Prompt

You are answering a longitudinal social survey as the described person.

Demographic profile: <profile_text>

Diversity-Oriented Prompting

MULTILINGUAL. MULTILINGUAL (Wang et al. 2025) follows the DIRECT setting but translates the complete survey prompt into four widely used languages: Chinese, Spanish, English, and Arabic. For each respondent-question pair, one of the four languages is selected uniformly at random using a deterministic hash of the global seed, wave index, respondent ID, and question variable.

ANTI-STEREOTYPE. ANTI-STEREOTYPE (Sivakumar et al. 2025) follows the PROFILE setting and appends an explicit instruction discouraging demographic stereotyping. This baseline evaluates whether explicit anti-stereotyping guidance can mitigate demographic overgeneralization.

Anti-Stereotype Instruction

Do not assume that one demographic attribute determines another. When information is missing, preserve uncertainty rather than filling it with stereotypes.

Non-Parametric Memory

SIMVBG. SIMVBG (Du et al. 2025) first uses the current backbone model to expand each respondent’s demographic profile into a comprehensive second-person background story, which remains fixed across all survey waves. Story generation uses a batch size of four, temperature 0, sampling disabled, and a maximum output length of 2,048 tokens.

The generation prompt includes the following instructions:

Background-Story Generation Prompt

You are a background story writer. Your task is to craft a comprehensive backstory for a person based on the information provided below. You must include every single data point from the original information, without exception.

This person’s information: <profile_text>

During evaluation, the generated story is inserted before the shared survey prompt:

SimVBG Evaluation Prompt

You are answering a longitudinal social survey as the person described in the background story.

Background story: <background_story>

FULL HISTORY. FULL HISTORY (Maharana et al. 2024) conditions the model on the demographic profile and all respondent-specific life events accumulated through the current wave:

Full-History Prompt

You are answering a longitudinal social survey as the described person.

Demographic profile: <profile_text>

Life history:

– <event_1>

– <event_2>

...

When the accumulated history exceeds the available input budget, events are retained in reverse chronological order under a character budget of $3 \times \text{max_input_length}$. Recent events are therefore preserved first, while older events may be truncated.

EVENT RAG. EVENT RAG (Lewis et al. 2020; Maharana et al. 2024) conditions the model on the demographic profile and the five accumulated life events most relevant to the current evaluation question. It uses bge-m3 to encode each evaluation question and each life-event text, where a life-event text concatenates the wave index, survey section, original question, answer text, and second-person statement.

For an event $e_{i,\tau}$ observed for respondent i at wave τ and a question q_t asked at wave t , the retrieval score is

$$s(q_t, e_{i,\tau}) = \text{sim}(q_t, e_{i,\tau}) \cdot \alpha_{\text{ret}}^{t-\tau}, \quad (16)$$

where $\alpha_{\text{ret}} = 0.9$ controls recency decay. The semantic similarity is computed as

$$\text{sim}(q_t, e_{i,\tau}) = \mathbf{h}_{q_t}^\top \mathbf{h}_{e_{i,\tau}}, \quad (17)$$

where $\mathbf{h}_{q_t}$ and $\mathbf{h}_{e_{i,\tau}}$ are ℓ_2 -normalized embeddings from bge-m3; hence the dot product is cosine similarity. The five highest-scoring events are inserted into the prompt.

Event RAG Prompt

You are answering a longitudinal social survey as the described person.

Demographic profile: <profile_text>

Relevant life experiences:

– <retrieved_event_1>

– <retrieved_event_2>

...

Setting	Value
LoRA rank	8
LoRA scaling coefficient	16
LoRA dropout	0
Target modules	q_proj, v_proj o_proj, down_proj
Learning rate	1×10^{-4}
Training epochs per update	2
Training batch size	16
Gradient accumulation steps	1
Maximum gradient norm	1.0

Table 7: Default configuration for agent-specific LoRA updates.

Control Baseline

RANDOM EVENT. RANDOM EVENT uses the same prompt structure and the same number of events as EVENT RAG. Instead of retrieving respondent-specific events, it samples five events from other agents using a pseudorandom generator initialized with the global simulation seed. This baseline preserves the amount and format of event information while removing the correspondence between the supplied events and the target respondent.

LifeMem Implementation Details

LifeMem combines demographic profile conditioning, structured life-event memory, and agent-specific parametric memory. At each survey wave, the system updates and evaluates every agent sequentially.

For each agent at every wave, LifeMem first loads the newly observed life events and updates the agent-specific LoRA adapter using both current and replayed events. It then adds the current events to structured memory, retrieves relevant events for each evaluation question, and constructs the corresponding prompt. After activating the agent-specific adapter, the model generates responses in batches, which are subsequently parsed and evaluated. Finally, the updated LoRA state is saved for use in the next wave.

Structured Life-Event Memory

LifeMem uses all-MiniLM-L6-v2 as a frozen retrieval encoder. Its lightweight architecture provides a practical balance between semantic retrieval quality and computational efficiency. Apart from the encoder choice, its retrieval procedure, including top- K , similarity scoring, and recency weighting, is identical to that of EVENT RAG. The recency term favors recent experiences while allowing older events to be retrieved when their semantic relevance remains high.

Agent-Specific Parametric Memory

Each agent maintains an independent LoRA adapter, while the parameters of the underlying language model remain frozen. Table 7 reports the default adapter and optimization settings.

Algorithm 1: Agent-specific parametric-memory update

```

1: for each wave  $t$  do
2:   for each agent  $i$  do
3:      $\mathcal{E}_{i,t}^{\text{cur}} \leftarrow$  events of agent  $i$  observed at wave  $t$ 
4:      $\mathcal{E}_{i,t}^{\text{rep}} \leftarrow$  sample  $R$  events from agent  $i$ 's previous events
5:      $\mathcal{B}_{i,t} \leftarrow \emptyset$ 
6:     for each event  $e \in \mathcal{E}_{i,t}^{\text{cur}}$  do
7:       Add  $(e.\text{question}, e.\text{answer})$  to  $\mathcal{B}_{i,t}$ 
8:       Add  $(e.\text{statement}, e.\text{first\_person})$  to  $\mathcal{B}_{i,t}$ 
9:     end for
10:    for each event  $e \in \mathcal{E}_{i,t}^{\text{rep}}$  do
11:      Add  $(e.\text{question}, e.\text{answer})$  to  $\mathcal{B}_{i,t}$ 
12:      Add  $(e.\text{statement}, e.\text{first\_person})$  to  $\mathcal{B}_{i,t}$ 
13:      Mark both examples with  $\text{replay} = \text{true}$ 
14:    end for
15:    Activate agent  $i$ 's LoRA adapter
16:    Save the pre-update parameters  $\Theta_{i,t}^{\text{before}} \leftarrow \Theta_{i,t}$ 
17:    for each training epoch do
18:      for each minibatch  $\mathcal{M} \subset \mathcal{B}_{i,t}$  do
19:        for each example  $z \in \mathcal{M}$  do
20:           $\ell_z \leftarrow \text{CE}(f_{\Theta_{i,t}}(z_{\text{input}}), z_{\text{assistant}})$ 
21:          if  $z$  is a replay example then
22:             $\ell_z \leftarrow \eta \ell_z$ 
23:          end if
24:        end for
25:         $\mathcal{L}_{\text{CE}} \leftarrow \frac{1}{|\mathcal{M}|} \sum_{z \in \mathcal{M}} \ell_z$ 
26:         $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \gamma \|\Theta_{i,t} - \Theta_{i,t}^{\text{before}}\|_2^2$ 
27:        Backpropagate  $\mathcal{L}$ 
28:        Clip gradients using  $\text{max\_grad\_norm}$ 
29:        Update the active adapter using the optimizer
30:      end for
31:    end for
32:    Save agent  $i$ 's updated LoRA adapter state
33:  end for
34: end for

```

Training examples. Each life event is converted into two training examples. The first teaches the association between the original survey question and the respondent's observed answer:

Question–Answer Training Example

User: <event_question>

Assistant: <event_answer_text>

The second reconstructs a first-person self-memory from the corresponding second-person event statement:

Self-Memory Reconstruction Example

User: Recall the following personal information: <second_person_event>

Assistant: <first_person_event>

The adapter therefore learns both the mapping from survey

questions to observed responses and the reconstruction of life events as first-person personal memories.

Sequential adapter updates. At each wave, LifeMem updates every agent's adapter using the events newly observed at that wave together with replayed events sampled from the same agent's history. Algorithm 1 summarizes the training procedure.

Replay events are sampled exclusively from the same agent's previously observed life events. By default, we uniformly sample up to $R = 4$ historical events and assign replay examples a loss weight of $\eta = 0.5$. Each replayed event yields the same question–answer and self-memory reconstruction examples as a current event. Sampling uses a pseudorandom generator initialized with the global simulation seed. When fewer than R historical events are available, all available events are included.

Stability regularization is enabled by default. Before updating an agent at wave t , we store the current adapter parameters $\Theta_{i,t}^{\text{before}}$ and add

$$\gamma \|\Theta_{i,t} - \Theta_{i,t}^{\text{before}}\|_2^2 \quad (18)$$

to the training objective, where $\gamma = 0.01$. This term limits abrupt parameter changes and reduces the extent to which newly observed events overwrite previously accumulated parametric memory.

During inference, multiple evaluation questions for the same agent are generated in batches while that agent's LoRA adapter remains active. This avoids repeatedly switching adapters between individual questions and ensures that all responses for an agent at a given wave use the same parametric state.

Evaluation Metrics

All metrics are computed only from valid human–model response pairs. Specifically, a sample is retained only when the human response corresponds to a predefined valid option and the model output can be parsed into a valid option from the same response space. Human responses coded as missing, refusal, “do not know,” or “not applicable” are excluded. This shared filtering rule prevents differences in missing-response behavior from affecting comparisons between methods.

Let $\mathcal{C}$ denote the set of eligible wave–question cells. For a cell $(t, q) \in \mathcal{C}$, let $\mathcal{Y}_{tq}$ be its set of valid response labels, and let $n_h^{tq}(y)$ and $n_m^{tq}(y)$ denote the numbers of valid human and model responses assigned to label y , respectively.

KL Divergence

For each wave–question cell, we construct empirical human and model response distributions. To avoid zero probabilities, we add $\epsilon = 10^{-9}$ to the count of every valid label:

$$P_h^{tq}(y) = \frac{n_h^{tq}(y) + \epsilon}{\sum_{y' \in \mathcal{Y}_{tq}} n_h^{tq}(y') + |\mathcal{Y}_{tq}| \epsilon}, \quad (19)$$

and

$$P_m^{tq}(y) = \frac{n_m^{tq}(y) + \epsilon}{\sum_{y' \in \mathcal{Y}_{tq}} n_m^{tq}(y') + |\mathcal{Y}_{tq}| \epsilon}. \quad (20)$$

Dataset	Grouping variables
Add Health	BIO_SEX, H1GI1Y, H1GI9, H1RE1
UKHLS	Sex, Birthy, Marstat, Employ, Oprlg1, Qfhigh, Jbstat

Table 8: Demographic variables used to define groups for the within-group pairwise distance metric. UKHLS denotes Understanding Society.

We then compute the KL divergence from the human distribution to the model distribution:

$$\text{KL}_{tq} = \text{KL}(P_h^{tq} \parallel P_m^{tq}) = \sum_{y \in \mathcal{Y}_{tq}} P_h^{tq}(y) \log \frac{P_h^{tq}(y)}{P_m^{tq}(y)}. \quad (21)$$

The reported KL divergence is the unweighted mean across all eligible wave-question cells:

$$\text{KL} = \frac{1}{|\mathcal{C}|} \sum_{(t,q) \in \mathcal{C}} \text{KL}_{tq}. \quad (22)$$

Lower values indicate closer alignment between the model and human response distributions.

Normalized Entropy Gap

For a response distribution P over the valid label set $\mathcal{Y}_{tq}$, normalized entropy is defined as

$$H(P) = -\frac{1}{\log |\mathcal{Y}_{tq}|} \sum_{y \in \mathcal{Y}_{tq}} P(y) \log P(y). \quad (23)$$

The normalization bounds entropy between zero and one for questions with at least two valid response options. We compute the human and model entropies separately:

$$H_h^{tq} = H(P_h^{tq}), \quad H_m^{tq} = H(P_m^{tq}). \quad (24)$$

The entropy gap for each wave-question cell is the absolute difference between them:

$$\text{EntropyGap}_{tq} = |H_m^{tq} - H_h^{tq}|. \quad (25)$$

The reported normalized entropy gap is

$$\text{NormEntropyGap} = \frac{1}{|\mathcal{C}|} \sum_{(t,q) \in \mathcal{C}} \text{EntropyGap}_{tq}. \quad (26)$$

A smaller value indicates that the overall concentration or dispersion of model responses is closer to that of human responses.

Within-Group Pairwise Distance Gap

The within-group pairwise distance measures whether response diversity within demographic groups is preserved. We form groups separately using each discrete demographic variable. The grouping variables are shown in Table 8.

Consider demographic variable g , group value v , wave t , and question q . Let $n_h^{tqgv}(y)$ be the number of human respondents in the group who select option y , and let

$$n_h^{tqgv} = \sum_{y \in \mathcal{Y}_{tq}} n_h^{tqgv}(y). \quad (27)$$

For groups containing at least two valid responses, the human within-group categorical pairwise distance is

$$D_h(t, q, g, v) = 1 - \frac{\sum_{y \in \mathcal{Y}_{tq}} n_h^{tqgv}(y) (n_h^{tqgv}(y) - 1)}{n_h^{tqgv} (n_h^{tqgv} - 1)}. \quad (28)$$

This quantity is the probability that two distinct respondents sampled from the same group provide different answers. The model distance is computed analogously:

$$D_m(t, q, g, v) = 1 - \frac{\sum_{y \in \mathcal{Y}_{tq}} n_m^{tqgv}(y) (n_m^{tqgv}(y) - 1)}{n_m^{tqgv} (n_m^{tqgv} - 1)}. \quad (29)$$

Group cells with fewer than two valid responses are excluded. Let $\mathcal{G}_{tq}$ denote the set of eligible demographic-variable-value cells for wave t and question q . We first average the group-specific distances:

$$D_h(t, q) = \frac{1}{|\mathcal{G}_{tq}|} \sum_{(g,v) \in \mathcal{G}_{tq}} D_h(t, q, g, v), \quad (30)$$

$$D_m(t, q) = \frac{1}{|\mathcal{G}_{tq}|} \sum_{(g,v) \in \mathcal{G}_{tq}} D_m(t, q, g, v). \quad (31)$$

The wave-question-level gap is then

$$\text{WGGap}_{tq} = |D_m(t, q) - D_h(t, q)|. \quad (32)$$

Finally, the reported within-group pairwise distance gap is

$$\text{WGPairwiseDistGap} = \frac{1}{|\mathcal{C}_{\text{WG}}|} \sum_{(t,q) \in \mathcal{C}_{\text{WG}}} \text{WGGap}_{tq}, \quad (33)$$

where $\mathcal{C}_{\text{WG}}$ contains wave-question cells with at least one eligible group cell. Lower values indicate closer alignment between model and human within-group response diversity.

Transition-Distribution JS Divergence

Transition-distribution JS divergence evaluates whether agents reproduce longitudinal patterns of response change rather than only the marginal response distribution at an individual wave. It is computed using the common longitudinal evaluation set described in Appendix .

For a common question q , respondent i , and two adjacent waves t and $t + 1$, the human response transition is

$$y_{i,t}^q \rightarrow y_{i,t+1}^q, \quad (34)$$

while the corresponding model transition is

$$\hat{y}_{i,t}^q \rightarrow \hat{y}_{i,t+1}^q. \quad (35)$$

Category	Method	Add Health			Understanding Society		
		KL Div. ↓	WG Gap ↓	Ent. Gap ↓	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
LLAMA-3.1-8B-INSTRUCT							
Static	Direct	14.678±0.192	0.551±0.009	0.698±0.009	15.317±0.144	0.552±0.004	0.698±0.008
Conditioning	Profile	9.147±0.450	0.389±0.006	0.512±0.004	7.234±0.072	0.376±0.006	0.486±0.011
Diversity-Oriented	Multilingual	10.459±0.109	0.368±0.007	0.497±0.006	11.653±0.074	0.341±0.004	0.472±0.004
	Anti-Stereotype	10.317±0.470	0.425±0.010	0.556±0.011	8.372±0.197	0.401±0.006	0.515±0.004
Prompting	SimVBG	7.540±0.639	0.343±0.004	0.452±0.015	5.950±0.315	0.345±0.008	0.454±0.017
	Full History	6.462±0.400	0.316±0.007	0.407±0.009	4.658±0.342	0.311±0.008	0.388±0.011
Non-Parametric Memory	Event RAG	6.086±0.307	0.291±0.008	0.388±0.011	5.068±0.209	0.308±0.011	0.393±0.008
	Random Event	7.399±0.496	0.347±0.005	0.455±0.002	5.443±0.125	0.341±0.003	0.446±0.007
Control Baseline	Random Event	7.399±0.496	0.347±0.005	0.455±0.002	5.443±0.125	0.341±0.003	0.446±0.007
Proposed Method	LifeMem	4.351±0.354	0.240±0.008	0.325±0.008	3.320±0.202	0.280±0.008	0.371±0.009
MINISTRAL-3-8B-INSTRUCT-2512							
Static	Direct	15.533±0.248	0.560±0.009	0.710±0.008	15.408±0.083	0.561±0.004	0.713±0.006
Conditioning	Profile	8.558±0.229	0.395±0.015	0.511±0.014	6.065±0.674	0.346±0.015	0.437±0.022
Diversity-Oriented	Multilingual	12.705±0.200	0.350±0.011	0.502±0.011	13.114±0.070	0.393±0.002	0.532±0.003
	Anti-Stereotype	9.842±0.355	0.425±0.009	0.542±0.011	8.243±0.501	0.381±0.012	0.490±0.014
Prompting	SimVBG	6.526±0.406	0.325±0.020	0.437±0.018	4.744±0.272	0.303±0.021	0.394±0.020
	Full History	5.271±0.550	0.303±0.013	0.391±0.009	3.462±0.623	0.259±0.041	0.321±0.033
Non-Parametric Memory	Event RAG	5.198±0.214	0.272±0.011	0.373±0.015	4.128±0.237	0.273±0.014	0.334±0.008
	Random Event	5.887±0.357	0.348±0.009	0.452±0.011	4.467±0.226	0.307±0.008	0.393±0.010
Control Baseline	Random Event	5.887±0.357	0.348±0.009	0.452±0.011	4.467±0.226	0.307±0.008	0.393±0.010
Proposed Method	LifeMem	2.164±0.243	0.195±0.005	0.272±0.010	1.850±0.144	0.207±0.020	0.251±0.014
QWEN3.5-9B							
Static	Direct	13.960±0.140	0.543±0.006	0.687±0.007	16.041±0.044	0.551±0.004	0.703±0.005
Conditioning	Profile	7.690±0.289	0.382±0.020	0.492±0.018	5.431±0.159	0.313±0.029	0.394±0.034
Diversity-Oriented	Multilingual	7.550±0.192	0.218±0.005	0.333±0.006	11.029±0.034	0.341±0.003	0.469±0.003
	Anti-Stereotype	7.374±0.418	0.355±0.021	0.469±0.012	6.323±0.348	0.312±0.026	0.400±0.024
Prompting	SimVBG	5.518±0.188	0.312±0.017	0.412±0.018	4.839±0.306	0.304±0.013	0.384±0.011
	Full History	4.486±0.091	0.254±0.005	0.341±0.002	3.488±0.251	0.273±0.024	0.329±0.014
Non-Parametric Memory	Event RAG	4.323±0.141	0.263±0.010	0.348±0.011	3.659±0.200	0.248±0.020	0.296±0.010
	Random Event	5.937±0.154	0.351±0.013	0.452±0.011	3.938±0.133	0.287±0.014	0.356±0.013
Control Baseline	Random Event	5.937±0.154	0.351±0.013	0.452±0.011	3.938±0.133	0.287±0.014	0.356±0.013
Proposed Method	LifeMem	4.015±0.259	0.266±0.009	0.354±0.010	2.622±0.140	0.228±0.021	0.276±0.016

Table 9: Robustness across respondent samples on Add Health and Understanding Society. Results report means across three independently sampled respondent sets generated with seeds 42, 43, and 44, with the corresponding $\pm$ standard deviations shown as subscripts. Each sample contains 100 respondents per dataset. KL Div., WG Gap, and Ent. Gap denote KL divergence, within-group pairwise distance gap, and normalized entropy gap. Lower values are better for all metrics. Bold marks the best mean result for each model and metric; LifeMem is shaded light gray.

We pool valid transitions across respondents and adjacent-wave pairs to construct empirical distributions over ordered response pairs:

$$P_h^q(a, b) = \Pr(y_t^q = a, y_{t+1}^q = b), \quad (36)$$

$$P_m^q(a, b) = \Pr(\hat{y}_t^q = a, \hat{y}_{t+1}^q = b), \quad (37)$$

where $(a, b) \in \mathcal{Y}_q \times \mathcal{Y}_q$ represents an ordered transition such as $1 \rightarrow 1$, $1 \rightarrow 2$, or $2 \rightarrow 1$. A transition is included only when both adjacent human responses are valid and both corresponding model responses can be parsed into valid options.

For each common question, we define the midpoint distribution

$$M^q = \frac{1}{2} (P_h^q + P_m^q), \quad (38)$$

and compute

$$\text{JS}_{\text{trans}}^q = \frac{1}{2} \text{KL}(P_h^q \| M^q) + \frac{1}{2} \text{KL}(P_m^q \| M^q). \quad (39)$$

The reported transition-distribution JS divergence is the mean across eligible common questions:

$$\text{TransitionJS} = \frac{1}{|\mathcal{Q}_{\text{common}}|} \sum_{q \in \mathcal{Q}_{\text{common}}} \text{JS}_{\text{trans}}^q. \quad (40)$$

Lower values indicate that the model more closely reproduces the human population-level distribution of response persistence and change across adjacent waves.

Additional Results

Robustness Across Respondent Samples

To evaluate robustness to respondent sampling, we repeat the main experiments using three independently sampled respondent sets generated with seeds 42, 43, and 44. Each run contains 100 respondents per dataset, while all other experimental settings remain unchanged. Table 9 reports the main-method comparison, while Table 10 reports the LifeMem

Method	Add Health			Understanding Society		
	KL Div. ↓	WG Gap ↓	Ent. Gap ↓	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
LLAMA-3.1-8B-INSTRUCT						
LifeMem (Full)	4.351 ± 0.354	0.240 ± 0.008	0.325 ± 0.008	3.320 ± 0.202	0.280 ± 0.008	0.371 ± 0.009
LifeMem w/o Param. Mem.	5.655 ± 0.302	0.266 ± 0.011	0.354 ± 0.012	4.744 ± 0.318	0.308 ± 0.016	0.402 ± 0.014
LifeMem w/o Struct. Mem.	6.161 ± 0.197	0.310 ± 0.003	0.413 ± 0.005	4.313 ± 0.413	0.317 ± 0.018	0.418 ± 0.013
MINISTRAL-3-8B-INSTRUCT-2512						
LifeMem (Full)	2.164 ± 0.243	0.195 ± 0.005	0.272 ± 0.010	1.850 ± 0.144	0.207 ± 0.020	0.251 ± 0.014
LifeMem w/o Param. Mem.	4.478 ± 0.210	0.246 ± 0.006	0.338 ± 0.006	3.680 ± 0.206	0.267 ± 0.012	0.329 ± 0.010
LifeMem w/o Struct. Mem.	2.981 ± 0.428	0.242 ± 0.028	0.327 ± 0.040	2.268 ± 0.152	0.236 ± 0.011	0.300 ± 0.012
QWEN3.5-9B						
LifeMem (Full)	4.015 ± 0.259	0.266 ± 0.009	0.354 ± 0.010	2.622 ± 0.140	0.228 ± 0.021	0.276 ± 0.016
LifeMem w/o Param. Mem.	4.424 ± 0.040	0.267 ± 0.010	0.356 ± 0.009	3.619 ± 0.247	0.253 ± 0.019	0.304 ± 0.014
LifeMem w/o Struct. Mem.	6.057 ± 0.435	0.348 ± 0.018	0.459 ± 0.017	3.521 ± 0.270	0.274 ± 0.013	0.342 ± 0.015

Table 10: Robustness of the LifeMem ablations across respondent samples. Results report means across three independently sampled respondent sets generated with seeds 42, 43, and 44, with the corresponding $\pm$ standard deviations shown as subscripts. KL Div., WG Gap, and Ent. Gap denote KL divergence, within-group pairwise distance gap, and normalized entropy gap. All metrics are lower-is-better. Bold marks the best mean result for each model and metric; LifeMem (Full) is shaded light gray.

ablation results. Both tables present the mean and standard deviation across the three respondent samples.

Table 9 shows that LifeMem remains the strongest method across respondent samples, achieving the best mean result in 16 of the 18 model–dataset–metric settings. The generally small standard deviations indicate that the main conclusions are not driven by a particular respondent sample.

Table 10 further shows that the full LifeMem model consistently outperforms both ablations. This result supports the complementary contributions of structured event memory and persistent parametric memory across different respondent samples.

Valid Response Rates

Table 11 shows that all LLM-based methods achieve high valid response rates across models and datasets, generally exceeding those in the human survey data. The lower human rates reflect refusals, nonresponse, and other missing-value cases that naturally occur in real-world surveys. The consistently high rates across methods also indicate that the experimental framework reliably produces responses that can be mapped to the predefined answer options.

Mechanism and Representation Analyses

LoRA Visualization Procedure

To visualize the evolution of agent-specific parametric memory, we extract the LoRA parameters of every agent after each survey wave. For agent i at wave t , all LoRA parameter tensors are flattened and concatenated into a single vector:

$$\theta_{i,t} = \text{concat}_{m \in \mathcal{M}} \text{vec}(\Theta_{i,t}^{(m)}), \quad (41)$$

where $\mathcal{M}$ denotes the consistently ordered set of LoRA tensors, including the corresponding `lora_A` and `lora_B`

parameters across adapted modules. Using a fixed ordering ensures that each dimension has the same parameter identity across agents and waves.

We stack the vectors from all N agents and T waves into a single matrix:

$$\mathbf{X} = \begin{bmatrix} \theta_{1,1}^\top \\ \theta_{1,2}^\top \\ \vdots \\ \theta_{N,T}^\top \end{bmatrix}. \quad (42)$$

Each parameter dimension is then standardized globally across all agent–wave states:

$$X'_{j,d} = \frac{X_{j,d} - \mu_d}{\sigma_d + \epsilon}, \quad (43)$$

where μ_d and σ_d are the mean and standard deviation of dimension d over all rows of $\mathbf{X}$, and ϵ is a small constant for numerical stability. This shared normalization prevents dimensions with larger numerical scales from dominating the projection.

We fit a single two-dimensional PCA projection to the globally standardized matrix:

$$\mathbf{Z} = \text{PCA}_2(\mathbf{X}'). \quad (44)$$

The resulting point $\mathbf{z}_{i,t} \in \mathbb{R}^2$ represents the LoRA state of agent i after wave t . For each agent, states are connected in chronological order,

$$\mathbf{z}_{i,1} \rightarrow \mathbf{z}_{i,2} \rightarrow \cdots \rightarrow \mathbf{z}_{i,T}, \quad (45)$$

forming a longitudinal parameter trajectory. Point colors indicate survey waves, while connecting lines identify states belonging to the same agent.

Method	Add Health			Understanding Society		
	Llama-3.1-8B	Ministral-3-8B	Qwen3.5-9B	Llama-3.1-8B	Ministral-3-8B	Qwen3.5-9B
Human	87.37%			64.31%		
Direct	97.51%	96.09%	97.51%	91.33%	90.87%	92.27%
Profile	97.45%	96.84%	97.51%	91.58%	91.14%	93.04%
Multilingual	92.87%	96.14%	97.51%	89.07%	91.01%	92.00%
Anti-Stereotype	97.45%	97.17%	97.51%	91.54%	90.97%	93.29%
SimVBG	97.40%	97.30%	97.49%	91.60%	91.71%	93.02%
Full History	96.84%	97.16%	97.51%	91.73%	91.35%	93.34%
Event RAG	97.24%	96.99%	97.45%	91.63%	91.04%	93.15%
Random Event	97.33%	97.18%	97.51%	91.72%	91.05%	93.11%
LifeMem	97.28%	96.41%	97.35%	92.28%	91.59%	93.34%

Table 11: Valid response rates across different methods and backbone models on Add Health and Understanding Society. A response is considered valid when it can be mapped to one of the predefined answer options. Higher values indicate fewer missing or unparseable responses.

α_{ret}	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
0.50	4.394	0.228	0.351
0.70	4.396	0.222	0.346
0.80	4.260	0.2201	0.342
0.90	4.063	0.231	0.321
0.95	4.487	0.2202	0.344
1.00	4.414	0.224	0.349

Table 12: Effect of the temporal decay coefficient α_{ret} in LifeMem on Add Health with Llama-3.1-8B-Instruct. Lower values are better for all metrics. Bold marks the best result for each metric, and the setting used in the main experiments is shaded light gray.

η	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
0.00	4.381	0.215	0.345
0.25	4.353	0.218	0.342
0.50	4.064	0.231	0.321
0.75	4.370	0.219	0.341
1.00	4.329	0.218	0.343

Table 13: Effect of the replay loss weight η in LifeMem on Add Health with Llama-3.1-8B-Instruct. Lower values are better for all metrics. Bold marks the best result for each metric, and the setting used in the main experiments is shaded light gray.

Additional LoRA Visualizations

Figures 8–12 extend the *Evolution of Agent-Specific LoRA States* analysis in the main paper, which presents only the Understanding Society results with Llama-3.1-8B-Instruct. Here, we provide the corresponding visualizations for all remaining model–dataset combinations: the three backbone models on Add Health and Ministral-3-8B-Instruct-2512 and Qwen3.5-9B on Understanding Society. Each point represents an agent’s LoRA state after a survey wave, while connected points trace the same agent over time. Together, these figures show whether the longitudinal evolution of agent-specific parametric states observed in the main-paper example also appears across other backbone models and datasets.

LoRA States across Life-Event Coverage

Figure 7 compares the LoRA trajectories obtained when each wave is allowed to use at most 5, 20, 40, or 90 life events.

With only a small number of life events, the adapter trajectories remain comparatively concentrated. As event availability increases, later-wave states show clearer agent-specific differentiation.

Hyperparameter and Design Analyses

Temporal Decay

We examine the temporal decay coefficient α_{ret} in LifeMem on Add Health with Llama-3.1-8B-Instruct. We vary α_{ret} over $\{0.5, 0.7, 0.8, 0.9, 0.95, 1.0\}$ while keeping all other settings fixed. A larger value assigns relatively more weight to earlier life events, whereas a smaller value emphasizes more recent events.

Table 12 suggests that an intermediate temporal decay coefficient provides the most favorable overall performance among the evaluated settings. In particular, $\alpha_{\text{ret}} = 0.9$ achieves the lowest KL divergence and entropy gap. Smaller values may discount earlier events too strongly, whereas values closer to 1 preserve older events with little decay. These results suggest that balancing recent and earlier experiences is useful for maintaining both distributional and diversity alignment.

Replay

We examine the replay loss weight η in LifeMem on Add Health with Llama-3.1-8B-Instruct. We vary η over $\{0, 0.25, 0.5, 0.75, 1.0\}$ while keeping all other settings fixed. A larger η places greater emphasis on replayed his-

Method	Retriever	Add Health			Understanding Society		
		KL Div. ↓	WG Gap ↓	Ent. Gap ↓	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
Event RAG	BM25	5.357	0.295	0.389	4.237	0.290	0.356
	all-MiniLM-L6-v2	4.650	0.252	0.346	3.845	0.281	0.339
	E5-large-v2	5.537	0.294	0.399	4.466	0.299	0.355
	BGE-M3	5.437	0.285	0.390	4.334	0.289	0.341
	BGE-M3 + BGE-Reranker-v2-M3	5.035	0.280	0.375	3.597	0.277	0.322
LifeMem	all-MiniLM-L6-v2	2.296	0.193	0.278	1.966	0.228	0.266

Table 14: Effect of retriever choice on EVENT RAG and LifeMem with Ministral-3-8B-Instruct-2512. Lower values are better for all metrics. Bold marks the best overall result, and LifeMem is shaded light gray.

r	KL Div. ↓	WG Gap ↓	Ent. Gap ↓
4	4.876	0.227	0.356
8	4.063	0.231	0.321
16	4.041	0.196	0.326
32	3.289	0.165	0.298

Table 15: Effect of the LoRA rank r in LifeMem on Add Health with Llama-3.1-8B-Instruct. Lower values are better for all metrics. Bold marks the best result for each metric, and the setting used in the main experiments is shaded light gray.

torical events during optimization, whereas a smaller η prioritizes examples from the current wave. The setting $\eta = 0$ disables the contribution of replay examples to the training loss.

Table 13 suggests that an intermediate replay weight provides a favorable trade-off across the evaluated metrics. In particular, $\eta = 0.5$ yields the lowest KL divergence and entropy gap, although smaller replay weights achieve a lower WG Gap. This pattern indicates that replay can support the retention of earlier experiences, but assigning it excessive or insufficient weight does not consistently improve all evaluation dimensions.

LoRA Rank

We examine the LoRA rank r in LifeMem on Add Health with Llama-3.1-8B-Instruct. We vary r over $\{4, 8, 16, 32\}$ while keeping all other settings fixed. A larger rank increases the capacity of the agent-specific adapter, but also raises its training and storage cost.

Table 15 shows that performance generally improves as the LoRA rank increases, with $r = 32$ achieving the lowest values across all three metrics. The improvement is especially clear for KL divergence and WG Gap, suggesting that additional adapter capacity may better support the representation of heterogeneous longitudinal information. We nevertheless use $r = 8$ in the main experiments because it provides a more practical balance between predictive performance, computational cost, and per-agent storage, which becomes important when simulating many agents across multiple waves.

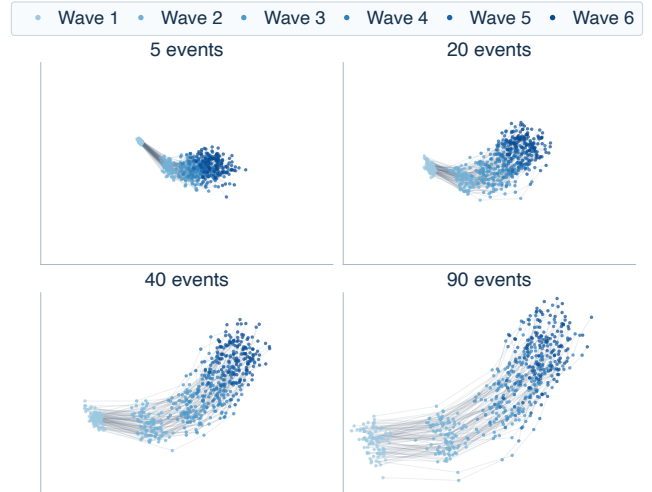


Figure 7: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Add Health with Llama-3.1-8B-Instruct under maximum life-event counts of 5, 20, 40, and 90. Points denote LifeMem agent states after each survey wave, and connected trajectories trace the same agent over time. With fewer events, different agents remain relatively concentrated; richer histories produce clearer agent-specific differentiation in later waves.

Retriever Choice

We compare several retrieval backends for EVENT RAG on Add Health and Understanding Society with Ministral-3-8B-Instruct-2512. The evaluated retrievers include BM25, all-MiniLM-L6-v2, E5-large-v2, BGE-M3, and BGE-M3 followed by BGE-Reranker-v2-M3. We additionally report LifeMem, which uses all-MiniLM-L6-v2 for structured event retrieval while maintaining an agent-specific parametric memory.

Table 14 shows that retriever choice affects EVENT RAG, but no single retriever performs best on both datasets. all-MiniLM-L6-v2 performs best among the EVENT RAG variants on Add Health, while BGE-M3 with reranking performs best on Understanding Society. LifeMem achieves lower values across all metrics while using the same all-MiniLM-L6-v2 retriever, suggesting that its gains are not solely due to retriever choice.

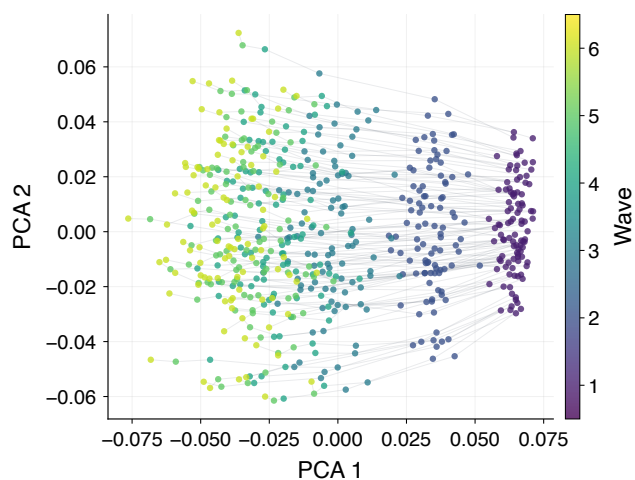


Figure 8: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Add Health with Llama-3.1-8B-Instruct. Points represent agent states after each survey wave, and connected trajectories trace the same agent over time. Colors indicate survey waves.

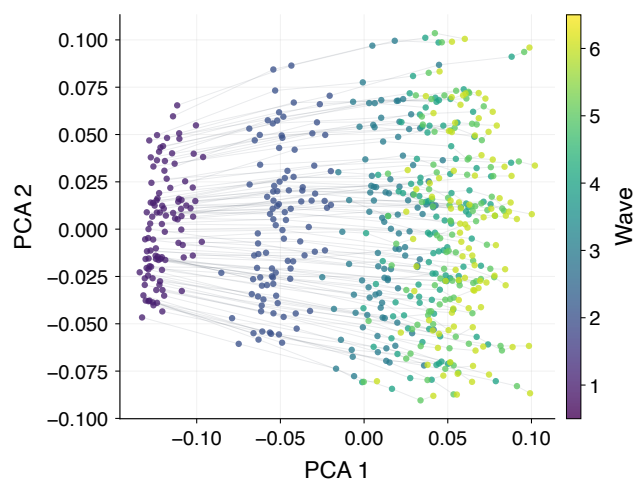


Figure 10: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Add Health with Qwen3.5-9B. Points represent agent states after each survey wave, and connected trajectories trace the same agent over time. Colors indicate survey waves.

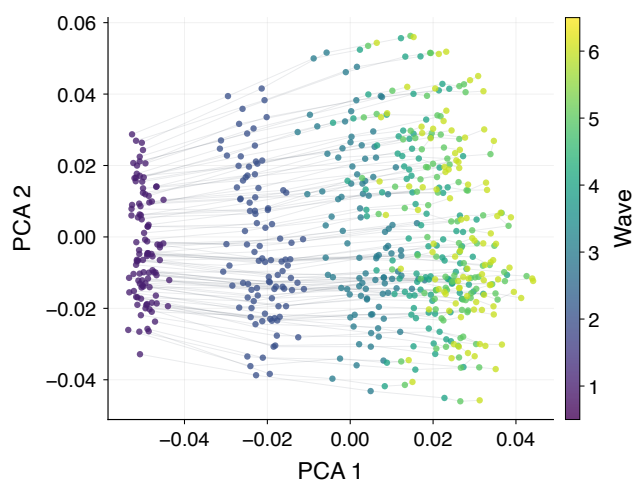


Figure 9: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Add Health with Ministral-3-8B-Instruct-2512. Points represent agent states after each survey wave, and connected trajectories trace the same agent over time. Colors indicate survey waves.

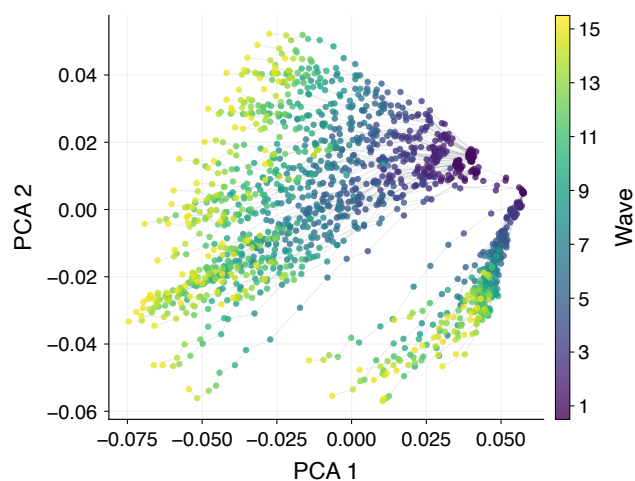


Figure 11: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Understanding Society with Ministral-3-8B-Instruct-2512. Points represent agent states after each survey wave, and connected trajectories trace the same agent over time. Colors indicate survey waves.

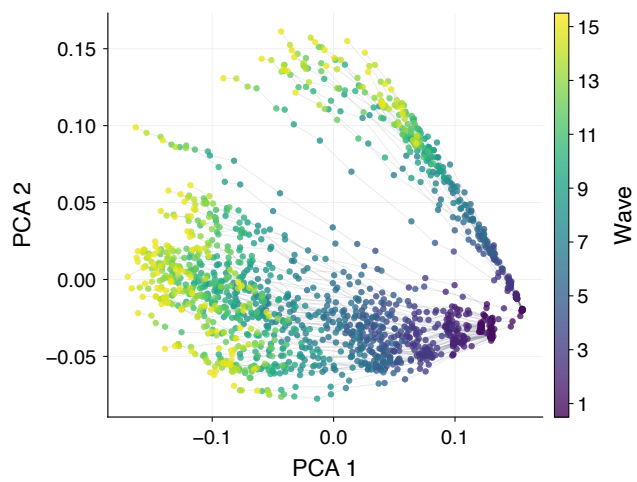


Figure 12: PCA visualization of agent-specific LoRA states learned by LifeMem for 100 agents on Understanding Society with Qwen3.5-9B. Points represent agent states after each survey wave, and connected trajectories trace the same agent over time. Colors indicate survey waves.